Rapid event-triggered adaptation of central pattern generator parameters for quadrupedal robot locomotion

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***Abstract*— Central pattern generators (CPGs) produce efficient rhythmic locomotion from a small set of parameters, but a set tuned for one terrain is no longer optimal once the terrain changes, even when a reactive attitude controller keeps the trunk level. We augment a coupled-Hopf-oscillator CPG with an attitude virtual model controller and study online re-tuning of its gait parameters as the terrain varies. We separate two questions: *when* to change the gait, resolved by an event trigger that fires on the controllable part of the prediction error, and *what* to change it to, where we compare grid search, Bayesian optimization, and an autoregressive active inference agent that minimizes expected free energy. On a natural transect of grass, gravel, rock, and shallow water, grid search and Bayesian optimization destabilize the robot while physically probing candidate gaits, raising the fall rate, whereas the active inference agent selects parameters from a learned model without such trials and is the only method to reduce falls below holding the incumbent gait. Generalizing this benefit across recurring terrain changes remains open.**

I. INTRODUCTION

Deploying legged machines in constantly changing environments demands an adaptability that engineers have long sought by replicating animal morphologies, such as snake-like robots for constrained spaces or quadruped platforms for irregular terrain [8], [4]. These platforms inherit the mechanical advantages of their biological counterparts, but their control systems often do not: the Boston Dynamics quadruped Spot, for instance, relies on centralized, sensor-intensive controllers that are effective yet complex and expensive, limiting wider deployment [4].

Central pattern generators (CPGs) offer a simpler alternative. In vertebrates, these spinal neural circuits generate the periodic, coordinated signals of movements such as walking, freeing higher brain centers to modulate speed and gait through simple inputs while sensory feedback keeps the rhythm coupled to the body and environment [5], [3]. This decentralized architecture reduces command latency and bandwidth and simplifies high-level control [6], making CPG-based controllers less energy intensive, simpler, and more affordable.

A. Related Work

Among CPG models used in robotics, coupled nonlinear oscillators are biologically plausible and computationally efficient [2], [6]. Their mutual inhibition synchronizes the joints into coordinated gaits with smooth transitions between them [2], [12], [10], and their limit-cycle behavior yields self-sustaining oscillations that recover quickly from perturbations [12]. Governed by only a few parameters such as frequency and amplitude, they suit embedded systems with limited resources, yet most implementations still target high-end platforms, leaving their use on compact systems underexplored.

An open-loop CPG generates stable rhythms on flat ground, but rough or sloped terrain induces body rotations that the rhythm alone cannot correct, so feedback is added to close the loop. Contact and load feedback resynchronize the oscillators to the actual footfalls [10], while posture feedback corrects the trunk attitude directly. Virtual model control (VMC) offers an intuitive route to the latter, computing joint actions as if virtual springs and dampers were attached to the body [9]. Ajallooeian et al. augment a quadruped CPG with virtual springs that pull the trunk level, together with a stumbling-correction reflex, and thereby traverse unperceived rough terrain and slopes that defeat an open-loop CPG [1]. We adopt this attitude VMC as a reactive posture layer beneath the CPG (Section II-C) and ask a complementary question: given such a controller, when and how should the CPG's own gait parameters be re-tuned as the terrain changes.

A complementary line of work tunes gait parameters as black-box optimization, from measured performance rather than a model of the dynamics. Bayesian optimization suits this setting, where each walking trial is expensive: it fits a Gaussian-process surrogate to the parameter-score pairs seen so far and proposes the next parameters by maximizing an acquisition function such as the upper confidence bound. Run *online*, it scores the parameters of the most recent window, augments the surrogate, and proposes the next window's parameters, refining the gait during walking; a trust region around the current best keeps proposals from jumping to destabilizing gaits [**?**]. We adopt this online Bayesian optimization as an adaptation baseline in Section IV.

In the following, we contribute

- the design of an intelligent agent that continuously updates parameters of a central pattern generator, and
- by benchmarking this agent against offline grid search and online Bayesian optimization.

II. PROBLEM STATEMENT

We first describe the base platform whose motion defines the locomotion objective, and then the controller driving it: a central pattern generator that produces the coordinated leg motion, augmented with an attitude virtual model controller

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that reactively corrects posture deviations. We then argue that, even with this reactive posture loop, the parameters yielding stable locomotion depend on the terrain, so that a parameter set tuned for one terrain degrades once the terrain changes. This motivates online adaptation of the parameters as a requirement for resuming stable locomotion.

A. Base platform and locomotion objective

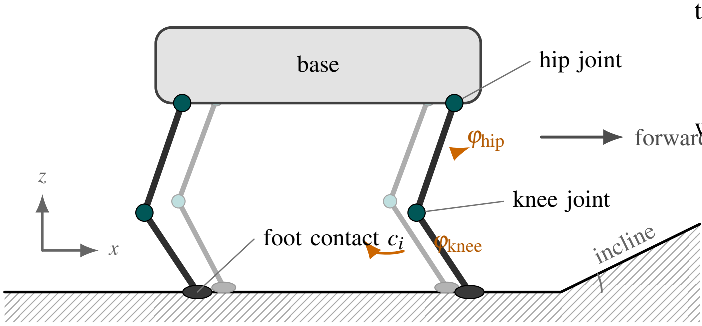

Fig. 1. The Laikago quadruped, its trunk base state ξ, and the twelve leg joints driven by the CPG.

The robot is a floating base actuated through its legs. We describe the trunk by its base state

$$\xi = (p, \Psi, v, \omega)\,, \tag{1}$$

collecting the linear position $p \in \mathbb{R}^3$ and the angular position (orientation) $\Psi \in SO(3)$ of the trunk in a world frame, together with the linear velocity $v = \dot{p}$ and the angular velocity ω. The orientation Ψ is parameterized by the roll, pitch, and yaw (ψ_r, ψ_p, ψ_y), of which the roll and pitch measure the deviation from an upright attitude.

The legs couple this base state to the joints. For a given motion of the base there is a corresponding configuration of the twelve joints, their rates, and the torques required to realize it,

$$(\varphi, \dot{\varphi}, \tau) = f(\xi)\,, \tag{2}$$

where φ are the joint angles, $\dot{\varphi}$ their rates, and τ the joint torques. The map f is unknown: it depends on the full rigid-body dynamics, the terrain, and the switching contact between the feet and the ground, none of which are available in closed form.

Quadrupedal locomotion is then the problem of finding the joint torques τ such that the base state follows a desired behavior. Concretely, we require the trunk to maintain a target forward velocity $v^\star$ while holding an upright angular position, that is roll and pitch near zero,

$$\tau^\star = \arg\min_{\tau} \left\| v - v^\star \right\| + \left\| (\psi_r, \psi_p) \right\|. \tag{3}$$

Because f is unknown, these torques cannot be solved for from a model directly. Instead, a central pattern generator produces rhythmic joint-angle references, an attitude virtual model controller adjusts these references to counter posture deviations, and joint-level position controllers track the result through torques, shaping the base motion toward the objective. We introduce both components next.

B. Central Pattern Generator

The CPGs are modeled as a network of four mutually coupled Hopf oscillators, each driving one limb: left front (LF), right front (RF), left hind (LH), and right hind (RH). Coordination between limbs arises from inhibitory coupling through a gait-specific coupling matrix $K = [k_{ij}]$. Let ω_i be the balance between stance and swing frequencies, operationalized as

$$\omega_i = \omega_{\text{stance}}\,\varsigma(-by_i) + \omega_{\text{swing}}\,\varsigma(by_i)\,, \tag{4}$$

with $\varsigma(\cdot)$ a sigmoid function. Then, the oscillators evolve as

$$\dot{x}_i = \alpha\,(u - r_i^2)\,x_i - \omega_i\,y_i\,, \tag{5}$$

$$\dot{y}_i = \beta\,(u - r_i^2)\,y_i + \omega_i x_i + \gamma \sum_{j=1}^{4} k_{ij}\,y_j + s_i\,, \tag{6}$$

where $r_i = \sqrt{x_i^2 + y_i^2}$ is the instantaneous amplitude, u defines the squared amplitude of the limit cycle, and $\alpha, \beta > 0$ are convergence rates (typically $\beta \gg \alpha$ to enforce stronger convergence along the y-axis). The term ω_i of Eq. 4 transitions between the stance and swing frequencies depending on the phase y_i: the sigmoid ς smoothly interpolates between the two frequencies according to the sign and magnitude of y_i, with the parameter b setting the sharpness of this transition. The coupling gain γ strengthens the coordination between the limbs and reduces the dependence on initial conditions; the coupling matrix $K = [k_{ij}]$ is gait-specific.

The term s_i closes a feedback loop on foot-contact timing [10]. Leg i is classified as swinging when y_i exceeds a small positive threshold and as in stance below a small negative one, with a hysteresis band in between. Comparing this phase against the binary contact state of foot i yields

$$s_i = \begin{cases} K_{\text{stop}}\big(\omega_i x_i - \gamma \sum_j k_{ij} y_j\big) & \text{contact matches phase} \\ F_{\text{fast}}\,\text{sign}(y_i) & \text{otherwise} \end{cases} \tag{7}$$

and $s_i = 0$ inside the hysteresis band. The contact matches the phase when the foot is airborne during swing or loaded during stance; it contradicts the phase upon an early touchdown or an unexpected loss of ground contact. The gains K_{stop} and F_{fast} set how strongly the sensed footfalls reshape the nominal rhythm, synchronizing the oscillators to the actual contact events.

The oscillator states map to joint-angle references,

$$\varphi_i^{\text{hip}} = \varphi_0^{\text{hip}} + A_{\text{hip}}\,x_i, \quad \varphi_i^{\text{knee}} = \varphi_0^{\text{knee}} - A_{\text{knee}}\max(0, y_i) + \Delta_i, \tag{8}$$

where $\varphi_0^{\text{hip}}, \varphi_0^{\text{knee}}$ are the joint offsets of the standing pose and $A_{\text{hip}}, A_{\text{knee}}$ are the swing amplitudes. During swing ($y_i > 0$) the knee flexes to lift the foot while the hip advances with x_i; during stance the knee holds its nominal extension. The term Δ_i is the posture correction supplied by the virtual model controller of Section II-C, and is zero for the open-loop CPG. Joint-level position controllers track these references, producing the joint torques τ of Eq. 2.

The behavior of the CPG is governed by a small set of parameters, collected in the vector

$$\theta = [\gamma, \omega_{\text{swing}}, \omega_{\text{stance}}, F_{\text{fast}}, K_{\text{stop}}, A_{\text{hip}}, A_{\text{knee}}, b], \quad (9)$$

comprising the coupling gain, the phase-dependent stance and swing frequencies, the fast-contact gain F_{fast} and stop gain K_{stop} of the contact feedback (Eq. 7), the hip and knee amplitudes of Eq. 8, and the sigmoid sharpness b of Eq. 4. These parameters set the gait shape, and are held fixed during locomotion.

C. Attitude Virtual Model Control

The contact feedback of Eq. 7 synchronizes the rhythm to the footfalls, but no part of the controller so far senses the trunk attitude. Yet roll and pitch deviations are precisely what drive the robot toward a fall (Eq. 3): on uneven or sloped ground the rhythmic pattern induces unwanted body rotations that it cannot itself correct. Virtual Model Control (VMC) addresses this by imagining virtual components, such as springs and dampers, attached to the robot and computing the joint actions that would arise if these components physically existed [9]. Ajallooeian et al. [1] augment a quadruped CPG with virtual springs spanning a trunk-fixed plane and a horizontal reference plane, so that the resulting virtual forces at the stance feet continually pull the trunk level.

We adopt this construction in a kinematic form suited to position-controlled joints. A virtual spring-damper on each attitude angle produces the correction commands

$$c_r = -\left(\kappa_p^r \psi_r + \kappa_d^r \dot{\psi}_r\right), \quad (10)$$

$$c_p = -\left(\kappa_p^p (\psi_p - \bar{\psi}_p) + \kappa_d^p \dot{\psi}_p\right), \quad (11)$$

where the κ's are stiffness and damping gains. The roll is regulated toward zero, since straight walking involves no sustained bank and lateral tip-over is the dominant fall mode. The pitch is regulated toward a slowly adapting baseline $\bar{\psi}_p$, an exponential moving average of the recent pitch with a time constant on the order of one second, so that transient rocking is damped while the sustained pitch offset imposed by an incline is tolerated rather than fought. The commands are distributed over the legs as the knee corrections of Eq. 8,

$$\Delta_i = \text{clip}\left(\sigma_i^F c_p + \sigma_i^L c_r, \pm\delta_{\max}\right), \quad (12)$$

where $\sigma_i^F = +1$ for the front legs and -1 for the hind legs, $\sigma_i^L = +1$ for the left legs and -1 for the right legs, and $\delta_{\max}$ bounds the correction. Adjusting a knee angle changes the effective leg length, so Eq. 12 realizes the virtual springs of Ajallooeian et al. [1] through the position-controlled knees: the legs on a sinking side of the trunk extend to push that corner back up, while the legs on the rising side shorten.

The VMC gains are held at fixed, hand-tuned values and are not part of the parameter vector θ. The attitude loop thus acts at a fast timescale, correcting momentary posture deviations within a stride, but it leaves the rhythmic pattern itself untouched. Ajallooeian et al. [1] observe that such momentary corrections lose effectiveness as the terrain becomes more difficult, when the attitude corrections begin to work against the forward progression planned by the CPG. Whether locomotion remains stable therefore still hinges on the gait parameters θ matching the terrain, which we examine next.

D. Terrain Dependence of the Optimal Parameters

For a fixed terrain, the parameter vector θ can be tuned to produce stable locomotion. Let e denote the terrain. Running the controller with parameters θ on terrain e, with the attitude feedback of Section II-C active, induces a distribution $p(\xi \mid \theta, e)$ over the base state, which we score by the locomotion objective of Eq. 3,

$$V(\theta, e) = -\mathbb{E}_{p(\xi \mid \theta, e)}\left[\frac{1}{v^\star}\left\|v - v^\star\right\| + \frac{1}{\psi_0}\left\|(\psi_r, \psi_p)\right\|\right], \quad (13)$$

setting $V = -c_f$ whenever the robot falls. The scale ψ_0 lets the attitude term dominate, encoding that remaining upright is the primary requirement. In practice the expectation is a time average over an evaluation window, computed on low-pass-filtered angles so that the gait's rhythmic rocking is not penalized, with the pitch measured relative to the terrain incline and progress beyond the target speed not rewarded. The best gait for terrain e is then the terrain-conditioned optimum

$$\theta^\star(e) = \arg\max_\theta V(\theta, e). \quad (14)$$

The difficulty is that $\theta^\star(e)$ is not constant across terrains. Optimizing Eq. 13 on surfaces that differ in incline, friction, and micro-geometry yields distinct parameter vectors: rougher or lower-friction ground calls for stronger inter-limb coupling, faster leg swing, and sharper stance-swing transitions than flat ground, and each parameter set is most stable on the terrain it was tuned for. On a natural landscape these properties drift continuously along the path, so the optimum $\theta^\star(e)$ drifts with them (Section IV).

Because θ is held fixed, a change in terrain leaves the controller at a now-suboptimal point $\theta^\star(e_{\text{old}}) \neq \theta^\star(e_{\text{new}})$, and the resulting attitude deviations compound: as the trunk tilts away from balance, recovery demands corrections beyond what the bounded attitude feedback of Eq. 12 can supply, and locomotion can fail before the new terrain is traversed. Resuming stable locomotion after a terrain change therefore requires detecting that the terrain has changed and re-tuning θ toward $\theta^\star(e_{\text{new}})$ online, quickly enough to preserve balance.

III. METHODS

A. Autoregressive active inference agent

The agent models the robot as a multivariate autoregressive process with exogenous inputs (MARX) [7]. The observation $y_k = (v_x, v_y, \psi_p, \psi_r)$ is a linear-Gaussian function of a regressor $x_k = [u_k,\ \bar{u}_k,\ \bar{y}_k]^\top$ that stacks the current control $u_k = \theta$ with short memories $\bar{u}_k, \bar{y}_k$ of past inputs and outputs,

$$p(y_k \mid \Theta, x_k) = \mathcal{N}\left(y_k \mid A^\top x_k, W^{-1}\right), \quad (15)$$

with parameters $\Theta = (A, W)$. A matrix-normal-Wishart prior is conjugate to this likelihood [11, ID: P323], so the belief over Θ is updated in closed form by rank-one Bayesian filtering as each pair (x_k, y_k) arrives, and marginalizing it gives a Student-t posterior predictive whose mean and covariance,

$$\mu_k(u) = M_{k-1}^\top x_k, \qquad \Sigma_k(u) = \frac{1 + x_k^\top \Lambda_{k-1}^{-1} x_k}{\nu_{k-1} - D_y - 1}\,\Omega_{k-1}, \quad (16)$$

are smooth functions of the control u through the regressor x_k. Here M, Λ, Ω, ν are the filtered posterior parameters and $D_y = \dim y$; we take the recursions as standard for this conjugate model [7], [11] and keep the focus on the control objective.

The agent selects controls by minimizing an expected free energy. Two priors close the model: a control prior $p(u) = \mathcal{N}(u \,|\, \mu_0, \Upsilon^{-1})$ regularizes the parameters toward a reference gait μ_0 with precision Υ, and a goal prior $\bar{p}(y) = \mathcal{N}(y \,|\, m_*, S_*)$ encodes the target observation, the desired forward velocity $v^\star$ at an upright attitude, with S_* setting how tightly each channel is enforced. Planning over a horizon $\tau = \{k+1, \ldots, k+T\}$ unrolls the model, substituting predictive means for the not-yet-observed outputs, and minimizes the cumulative divergence of the predicted observations from the goal prior,

$$G(u_\tau) = \sum_{t \in \tau} D_{\mathrm{KL}}\big[\, p(y_t \,|\, u_t, \mathcal{D}_k) \,\big\|\, \bar{p}(y_t) \,\big]. \quad (17)$$

Expanding the divergence and adding the control prior gives the objective minimized at each planning step,

$$\begin{aligned} \hat{u}_\tau = \arg\min_{u_\tau} \sum_{t \in \tau} \Big[& \underbrace{\tfrac{1}{2}\operatorname{tr}\big(S_*^{-1}\Sigma_t(u_t)\big) - \tfrac{1}{2}\ln|\Sigma_t(u_t)|}_{\text{information gain}} \\ & + \underbrace{\tfrac{1}{2}\big(\mu_t(u_t) - m_*\big)^\top S_*^{-1}\big(\mu_t(u_t) - m_*\big)}_{\text{goal seeking}} \\ & + \underbrace{\tfrac{1}{2}\big(u_t - \mu_0\big)^\top \Upsilon \big(u_t - \mu_0\big)}_{\text{control cost}} \Big]. \end{aligned} \quad (18)$$

The information-gain terms, through the predictive covariance Σ_t, drive the agent to resolve model uncertainty; the goal-seeking term, a Mahalanobis distance of the predicted mean μ_t from the target m_*, drives it toward stable walking; and the control cost keeps it near the reference gait.

B. Prediction-error triggering

Re-tuning θ at every step is wasteful and, as Section IV shows, actively destabilizing, so the agent adapts only when the incumbent gait is no longer matched to the terrain. We detect that event by testing, online, whether the incumbent controls still minimize the control objective of Eq. 18.

Consider the one-step objective $G(u)$ under two simplifications that hold in the monitoring regime, once the gait is steady and the belief identified. The control cost is a fixed price rather than a signal, and the information-gain term is flat in u (its only dependence is through $\ln(1 + x^\top \Lambda^{-1} x)$, negligible for large Λ). What remains is the goal cross-entropy

$$G(u) = \tfrac{1}{2}\operatorname{tr}\big(S_*^{-1}\Sigma\big) + \tfrac{1}{2}\big(\mu(u) - m_*\big)^\top S_*^{-1}\big(\mu(u) - m_*\big), \quad (19)$$

with $\mu(u) = M^\top x$ affine in u through the learned control-to-output gain $B := \partial\mu/\partial u$, a block of $M^\top$. The incumbent θ remains terrain-matched precisely when it is a stationary point of G. Because μ is affine in u and G is quadratic in μ, G is exactly convex-quadratic in u, with gradient and Hessian

$$g = \nabla_u G(\theta) = B^\top S_*^{-1}(\mu - m_*), \quad H = \nabla_u^2 G = B^\top S_*^{-1} B. \quad (20)$$

The value $G(\theta)$ alone does not test optimality: the goal error $\mu - m_*$ can grow along directions the controls cannot affect, terrain error that B annihilates, leaving θ stationary despite a high cross-entropy. The gradient separates the fixable error from the rest. Its affine-invariant magnitude is the Newton decrement

$$\lambda^2 = g^\top H^{-1} g = \big\|\Pi_{\mathcal{R}(B)}(\mu - m_*)\big\|_{S_*^{-1}}^2, \quad (21)$$

the squared goal error projected onto the controllable subspace $\mathcal{R}(B)$. Since G is exactly quadratic, $\frac{1}{2}\lambda^2 = G(\theta) - \min_u G(u)$ is not a local approximation but the exact suboptimality gap, the cross-entropy a full re-tuning of the gait would recover. Triggering on λ^2 therefore fires on *fixable* mismatch rather than on raw prediction error, and correctly stays silent when the error lies in the uncontrollable subspace, where no gait switch would help.

The threshold follows from the same statistic. Under the null hypothesis that the incumbent is still optimal, the goal error is predictive noise, $\mu - m_* \sim \mathcal{N}(0, \Sigma)$, so λ^2 is a quadratic form of a Gaussian and hence (generalized) χ^2 with $r = \operatorname{rank}(B) \le \min(D_u, D_y)$ degrees of freedom. A target false-alarm rate α then fixes

$$K = \chi_r^2(1 - \alpha), \quad (22)$$

a threshold with units and a calibrated meaning in place of a hand-tuned level. Equivalently, restoring the control cost as the price of adapting gives a decision-theoretic rule: trigger only when the recoverable gain outweighs the cost of the corrective step $\Delta u = -H^{-1} g$, that is $\frac{1}{2}\lambda^2 > \frac{1}{2}\Delta u^\top \Upsilon \Delta u$, so that adaptation fires exactly when the full expected free energy would decrease.

In practice, calibrating the null scale of λ^2 on a short flat-walking prefix reduces the test to a dimensionless ratio near unity during nominal locomotion, the form used in Section IV. The attitude channels of S_* are tightened relative to the velocity channels so the statistic responds to a loss of balance rather than to a steady change in pitch, and it is smoothed by an exponential moving average to reject stride-frequency ripple. An event fires when the smoothed statistic exceeds K for P consecutive steps; the monitor then disarms and re-arms only once it falls back well below K, so a single terrain change produces one event while a corridor of repeated changes re-triggers once per transition. Symmetrically, adaptation is paused whenever the statistic returns below K, so re-tuning runs only while the incumbent is provably suboptimal.

IV. EXPERIMENTS

We evaluate three parameter-selection strategies for the Hopf-oscillator CPG controller on the Laikago quadruped in PyBullet: (i) an offline *grid search* over a small set of CPG parameters, a fixed non-adaptive reference; (ii) online *Bayesian optimization* (BO) with a Gaussian-process surrogate and a UCB acquisition function, a state-of-the-art black-box optimizer; and (iii) the proposed *MARX-EFE* active inference agent, which continually updates the parameters by minimizing expected free energy. All three search the same 8-dimensional space θ of Eq. 9 with identical bounds, and all run with the attitude VMC of Section II-C active. Control runs at $\Delta t = 0.01\,\mathrm{s}$ with a target forward velocity of 1.0 m/s; each candidate is scored over a steady-state window with the settling transient excluded, and parameter changes are interpolated over the first 150 steps so the gait never jumps.

A. Adaptation protocol

Adaptation is gated by the optimality monitor of Section III-B. A MARX belief (Eq. 15) is filtered online from the observation $y_k = (v_x, v_y, \psi_p, \psi_r)$ with forgetting factor 0.995, under a stability-oriented goal prior whose attitude channels are tightened, so the statistic responds to a loss of balance rather than to a steady pitch offset. At each step the monitor evaluates the Newton decrement λ^2 (Eq. 21), the goal cross-entropy that re-tuning the gait could recover, from the model's steady-state control gain B. Because B is the multi-step feedthrough of the identified model, λ^2 reflects the sustained mismatch a terrain change creates, whereas a one-step innovation at 100 Hz is dominated by persistence and barely registers it. An event fires when λ^2 exceeds the decision-theoretic budget of Section III-B, and the monitor re-arms once λ^2 falls back below half the budget, so a sequence of changes triggers once per transition rather than latching.

When the trigger fires, the selected method proposes CPG parameters, one candidate per 1.5 s window ramped in over 0.3 s, and each window is scored by the criterion of Eq. 13. The optimizers search a trust region around the incumbent and share a common safety layer: the best-scoring parameters seen so far are restored for one window whenever a candidate scores well below them. Without this safeguard every method, MARX-EFE included, destabilizes the robot within seconds, so it is applied uniformly. Grid search proposes candidates from a fixed sub-grid and BO from a Gaussian-process surrogate with a UCB acquisition function; both must physically walk on each candidate to score it, whereas MARX-EFE selects from its model, keeping its control prior centered on the current parameters. Holding the incumbent $\theta^\star(\text{flat})$ without adapting is the reference against which all three are measured.

B. Experiment 1: Natural terrain

The parameter optimum drifts continuously along a natural landscape, whose ground changes in both friction and geometry many times, so there is no single terrain-matched gait to switch to and no oracle to mark an achievable ceiling. We walk the robot along a natural transect of forward bands of grass, gravel, rock, and shallow water in random order and length, each band carrying its own friction and its own micro-geometry: gentle undulation on grass, fine roughness on gravel, scattered low rocks to step over, and a slippery depression at the water. Bands are 6 to 10 m long, so each surface persists for 12 to 20 s of walking, and the robot walks a single continuous bout of up to 80 s. It starts on grass with $\theta^\star(\text{flat})$, whose friction $\mu \approx 0.7$ makes the flat optimum the natural incumbent, and adapts online whenever the monitor fires, on average 1.5 times over the bout. Because there is no oracle, and almost every gait eventually falls on a long enough transect, the discriminating metric is the *survival distance*, the distance walked before a fall or the end of the terrain. We report it alongside the fall rate, the number of bands crossed, the mean trunk tilt $\sqrt{\psi_r^2 + \psi_p^2}$ after the baseline window, and the mean windowed criterion $\bar{V}$, over $N = 10$ sampled transects.

TABLE I
ONLINE ADAPTATION ON A NATURAL TRANSECT OF FRICTION AND GEOMETRY BANDS ($N = 10$ SAMPLED TRANSECTS, DECISION-THEORETIC TRIGGER). THERE IS NO ORACLE, BECAUSE THE OPTIMUM DRIFTS ALONG THE PATH. SURVIVAL DISTANCE AND BANDS CROSSED ARE HIGHER-IS-BETTER; FALL RATE AND TRUNK TILT ARE LOWER-IS-BETTER; $\bar{V}$ IS HIGHER-IS-BETTER. BEST ONLINE METHOD IN BOLD.

	reference	online method		
Metric	no adapt	grid	BO	MARX-EFE
Distance [m]	34.2	32.9	33.2	**34.3**
Falls [%]	50	60	60	**40**
Bands crossed	4.7	4.5	4.5	4.6
Trunk tilt [deg]	2.2	2.3	2.3	**2.2**
Mean $\bar{V}$	−0.27	−0.50	−0.43	**−0.31**

Table I reports the outcome. Survival distances overlap across all arms, from 33 to 34 m with a per-seed spread of about 11 m that reflects where the first fall happens, so distance alone does not separate the methods. The fall rate and the windowed criterion do, and they order the arms consistently. Online black-box search is the most fragile: grid and BO fall in 60% of runs against 50% for no adaptation, and their mean criterion is the lowest (−0.50 and −0.43 against −0.27), because a speculative candidate walked on a low-friction or uneven band tips the robot over. MARX-EFE is the only arm to improve on holding the incumbent, falling in 40% of runs while matching no adaptation on trunk tilt and criterion. It selects small model-guided corrections rather than committing whole windows to physical trials, which avoids the destabilization that black-box search causes. The small sample leaves these orderings short of statistical significance, but their direction is consistent and holds once the bands are long enough to expose a mismatched gait for a sustained stretch.

C. Experiment 2: Sensitivity to exploration schedule

A known failure mode of BO is that its behaviour depends strongly on the acquisition-function exploration schedule.

We sweep five β_t schedules (`conservative`, `moderate`, `balanced`, `late_decay`, `exploratory`) while holding all other BO settings fixed, and compare them against a single MARX-EFE run. The purpose is to show (a) that BO requires schedule tuning to achieve competitive sample efficiency, and (b) that MARX-EFE does not require such a schedule because the exploration–exploitation trade-off is implicit in the epistemic term of the expected free energy.

D. Experiment 3: Online adaptation to target-velocity changes

A key claim of the continually-learning agent is that it can adapt online to non-stationary tasks without re-tuning. We change the target velocity v^* mid-episode from 0.5 m/s to 1.0 m/s to 0.8 m/s, and measure how quickly each method tracks the new set-point. Only methods that update their parameters online (BO and MARX-EFE) are evaluated here.

V. DISCUSSION

Our results separate two aspects of online adaptation that are easily conflated: deciding *when* to change the gait and deciding *what* to change it to. The optimality trigger resolves the first cheaply and reliably, firing on the controllable part of the prediction error, once per terrain change and not on a steady surface, and it re-arms to detect a sequence of transitions. The second is where the methods diverge. Grid search and Bayesian optimization can only judge a candidate gait by walking on it, and on a low-friction or uneven band a poor candidate does not merely score badly, it tips the robot over; even confined to a trust region and protected by a revert-to-best safeguard, this exploration raises the fall rate relative to not adapting at all. The active inference agent instead selects parameters from a model that it updates every control step, so it avoids these destabilizing trials and is the only method to fall less often than holding the incumbent gait. It achieves this not by a faster search toward a distant optimum, but by making small, model-guided corrections around the incumbent: the advantage is the ability to improve the gait without risking it. This holds even though the CPG is already stabilized by the attitude virtual model controller, so the benefit is complementary to the reactive posture loop rather than a substitute for it.

Several limitations qualify these conclusions. The natural-terrain sample is small ($N = 10$), so the ordering is consistent but short of significance, and the reference optimum is computed offline in simulation. The improvement does not yet generalize to a durable advantage across recurring terrain changes: the selected parameters stay near the incumbent and never travel to a distant terrain optimum. The barrier appears structural rather than a matter of tuning: the linear autoregressive model supports small local corrections but cannot predict which distant parameter settings are stable, and loosening the control prior to permit larger moves only reintroduces destabilizing exploration. A model expressive enough to navigate to the terrain optimum, or a per-transition re-selection scheme, is the central open problem. Validating on hardware, and beyond the terrain envelope in which the attitude-stabilized CPG stays upright, are natural next steps.

VI. CONCLUSIONS

We augmented a coupled-Hopf-oscillator CPG with an attitude virtual model controller and studied when and how to re-tune its gait parameters online as the terrain varies. An optimality trigger decides when to adapt from the controllable part of the prediction error, and an autoregressive active inference agent decides what to change by selecting parameters from a learned model rather than by physically trialing candidates. On a natural transect it is the only method to reduce falls below holding the incumbent gait, where grid search and Bayesian optimization instead destabilize the robot while probing. Generalizing this benefit across recurring terrain changes, and validating on hardware, remain open.

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